

Testosterone Cypionate — Intramuscular Injection Instructions

PROVISIONAL CLINICAL CONTENT — This protocol has been drafted using current evidence-based parameters and is operationally usable upon Provisional approval. Full clinical review and final approval by the Covering Physician are scheduled within 180 days. If you have any questions about the technique described in this document, contact your Bloom Metabolics care team via the patient portal before injecting.

Protocol ID: Protocol-001

Medication: Testosterone Cypionate

Route: Intramuscular (IM)

Version: 0.1

Effective Date: TBD (pending approval)

Reviewed by: Pending — Dr. Michael Napolitano, MD

Before You Begin

Gather the following supplies before starting:

- & Testosterone cypionate vial (check expiration date — do not use if expired)
- & Intramuscular syringe: 23-gauge or 25-gauge, 1-inch needle, 1mL or 3mL barrel
- & Draw needle: 18-gauge (for drawing medication from the vial — do not inject with this needle)
- & Alcohol swabs (2 — one for the vial top, one for the injection site)
- & Sterile gauze pad
- & FDA-cleared sharps container
- & Clean, flat surface with good lighting

Before every injection:

1. Wash your hands thoroughly with soap and water for at least 20 seconds
2. Inspect the vial — the oil should be clear to pale yellow with no visible particles. If the solution is cloudy, discolored, or contains particles, do not use it. Contact your care team.
3. Check the expiration date on the vial
4. Warm the vial between your palms for 1–2 minutes — testosterone cypionate is oil-based and injects more smoothly when warmed to body temperature

If something does not look right — the vial is damaged, the solution looks unusual, or the needle packaging is open — do not inject. Contact your care team.

Identifying Your Syringe

This protocol uses an intramuscular syringe. This is a larger syringe (1mL or 3mL barrel) with a detachable needle.

Feature	Intramuscular Syringe (USE THIS)	Insulin Syringe (DO NOT USE)
Barrel size	1mL or 3mL	0.5mL or 1mL
Needle	Detachable, 23g–25g, 1 inch	Fixed (non-removable), 29g–31g, short
Needle thickness	Thicker (you can feel it is sturdier)	Very thin and flexible
Typical use	Intramuscular injections (into muscle)	Subcutaneous injections (into fat tissue)

If your shipment contains only insulin syringes (thin, testosterone with them. Contact your care team immediately if you have insulin syringes in your shipment. Insulin syringes are not appropriate for intramuscular injections.

You will use two needles:

1. Draw needle (18-gauge): A larger needle used only to draw medication from the vial. The wider gauge allows the oil to flow easily into the syringe. Do not inject with this needle.
2. Injection needle (23-gauge or 25-gauge, 1 inch): Swap to this needle after drawing. This is the needle you inject with.

Injection Sites

Recommended Sites

Ventrogluteal (Preferred)

The upper lateral hip, between the iliac crest and the greater trochanter. This is the preferred intramuscular injection site because it has thick muscle mass with minimal nerves and blood vessels.

How to locate: Place the palm of your hand on the greater trochanter (the bony prominence at the top of your thigh). Point your index finger toward the anterior iliac spine (the front of your hip bone) and spread your middle finger posteriorly. The injection site is in the V formed between your two fingers.

Vastus Lateralis (Preferred)

The outer middle third of the thigh, between the knee and the hip. This is an easy self-injection site because you can see and reach it without assistance.

How to locate: Divide your thigh into thirds (top, middle, bottom). The injection site is in the outer middle third — the area on the outside of your thigh roughly halfway between your knee and your hip.

Acceptable Sites

Deltoid

The lateral deltoid muscle, approximately 2–3 fingerbreadths below the acromion process (the bony point of your shoulder). Acceptable for injection volumes of 1mL or less only. The deltoid is a smaller muscle and cannot accommodate larger volumes.

Dorsogluteal

The upper outer quadrant of the buttock. Acceptable but use caution — this site has proximity to the sciatic nerve and requires careful landmarking. The ventrogluteal site is preferred over the dorsogluteal.

Site Rotation Schedule

Rotate your injection site with each administration. Alternate between left and right sides weekly. Cycle through recommended sites to allow tissue recovery.

Example 8-week rotation: Intramuscular Injection Instructions

Week	Site	Side	Date	Notes
1	Ventrogluteal	Left	_____	_____
2	Vastus Lateralis	Right	_____	_____
3	Ventrogluteal	Right	_____	_____
4	Vastus Lateralis	Left	_____	_____
5	Ventrogluteal	Left	_____	_____
6	Vastus Lateralis	Right	_____	_____
7	Ventrogluteal	Right	_____	_____
8	Vastus Lateralis	Left	_____	_____

Fill in the date and any observations (your follow-up appointments).

Step-by-Step Injection Technique

Drawing Medication

1. Clean the top of the testosterone vial with an alcohol swab. Allow it to air dry.
2. Attach the draw needle (18-gauge) to the syringe.
3. Pull back the plunger to draw air into the syringe equal to the volume of your prescribed dose.
4. Insert the draw needle into the vial through the rubber stopper.
5. Inject the air into the vial (this equalizes pressure and makes drawing easier).
6. Invert the vial and slowly pull back the plunger to draw your prescribed dose into the syringe.
7. Check for air bubbles. If present, tap the syringe gently with your finger to move bubbles to the top, then push the plunger slightly to expel the air. Re-check the volume.
8. Remove the needle from the vial.
9. Swap to the injection needle (23g or 25g, 1 inch). Remove the draw needle carefully and attach the injection needle. Do not touch the needle itself.

Injecting

1. Clean the injection site with a fresh alcohol swab using a circular motion from center outward. Allow to air dry completely.
2. With your non-dominant hand, spread the skin taut at the injection site (Z-track technique: pull the skin to the side approximately 1 inch before inserting the needle — this helps seal the medication in the muscle after withdrawal).
3. Hold the syringe like a dart with your dominant hand.
4. Insert the needle at a 90-degree angle to the skin surface in one smooth, quick motion. Insert the needle to its full 1-inch depth.
5. Inject the medication slowly and steadily. Take approximately 10 seconds per mL to reduce discomfort and ensure proper tissue absorption.
6. After injecting the full dose, wait 10 seconds before withdrawing the needle. This allows the medication to begin absorbing and reduces leakback.
7. Withdraw the needle in one smooth motion at the same angle it was inserted.
8. Release the Z-track skin. The displaced tissue seals over the injection track.

Testosterone Cypionate. Apply gentle pressure with a sterile gauze pad. Do not rub the site.

After Injection

1. Immediately dispose of the used syringe and both needles in your sharps container. Do not recap needles.
2. Record the injection in your rotation tracker (site, side, date, and any observations).

What to Expect After Injection

Normal findings (not cause for concern):

- Mild soreness or aching at the injection site for 24–72 hours
- A small drop of blood at the injection site after needle withdrawal
- Slight firmness or a small lump at the injection site that resolves within a few days
- Temporary mild redness at the puncture site

Abnormal findings (contact your care team):

- Increasing redness, warmth, or swelling at the injection site after 48 hours
- Drainage or pus from the injection site
- Fever (temperature above 100.4°F / 38°C) following injection
- Persistent pain that worsens rather than improves after 72 hours
- Numbness, tingling, or weakness in the leg or hip following injection (may indicate nerve involvement)

Troubleshooting

I saw blood when I pulled back on the plunger or after removing the needle.

A small amount of blood at the injection site is normal and results from puncturing superficial capillaries. Apply gentle pressure with a clean gauze pad. If you aspirated and drew blood into the barrel, withdraw the needle, discard the syringe, and prepare a new injection at a different site.

The needle bent during injection.

Do not attempt to straighten or reinsert a bent needle. Withdraw it carefully, discard it in your sharps container, and prepare a new syringe with a fresh needle. Ensure you are relaxing the target muscle before insertion.

I missed my scheduled injection day.

Administer your injection as soon as you remember. If it is close to your next scheduled dose, contact your care team for guidance. Do not double your dose.

The injection site is red, swollen, or painful 24+ hours later.

Mild soreness for 24–48 hours is common with intramuscular injections. Apply a warm compress to the area. If you observe increasing redness, warmth, swelling, drainage, or develop fever, contact your care team immediately.

I am not sure I injected the full dose.

If the syringe barrel appears empty, you likely received the full dose. A small amount of medication may remain in the needle hub — this is normal and accounted for in dosing. Do not attempt a second injection. Note this at your next check-in.

The oil was difficult to push through the needle.

Testosterone cypionate is an oil-based solution and may feel viscous. Warm the vial between your palms for 1–2 minutes before drawing. Inject slowly and steadily. Using a 23g needle rather than 25g may reduce resistance.

When to Contact Your Care Team

Contact your Bloom Metabolics care team via the OptiMantra patient portal if you experience any of the following:

- Injection site signs of infection (increasing redness, warmth, swelling, drainage, fever)
- Persistent pain worsening after 72 hours
- Numbness, tingling, or weakness after injection
- Allergic reaction symptoms (hives, swelling of face/lips/tongue, difficulty breathing)
- Uncertainty about your technique — contact us BEFORE injecting
- Any situation not covered by this troubleshooting guide

Emergency: If you are experiencing a medical emergency — difficulty breathing, severe allergic reaction, loss of consciousness — call 911 immediately. Do not delay emergency care to contact your clinic.

Storage and Disposal

Medication Storage

- Store testosterone cypionate at controlled room temperature: 68–77°F (20–25°C)
- Protect from light — store in the original packaging or a dark, dry location
- Do not refrigerate or freeze
- Do not use if the solution is cloudy, discolored, or contains visible particles
- Do not use past the expiration date on the vial

Sharps Disposal

- Dispose of all used needles and syringes immediately in an FDA-cleared sharps container
- Do not recap, bend, or break needles before disposal
- When the sharps container is three-quarters full, seal it and follow your local community disposal guidelines
- Many pharmacies accept sharps containers for disposal — contact your local pharmacy
- Do not dispose of loose needles in household trash

Unused Medication

- Do not flush testosterone down the toilet
 - Return unused or expired medication to your pharmacy or follow local medication disposal guidelines
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Acknowledgment

Before your first self-injection, you will acknowledge receipt and understanding of these instructions in the OptiMantra patient portal. The acknowledgment statement reads:

"I have reviewed the injection instructions for testosterone cypionate. I understand the technique, the injection sites, the rotation schedule, and the signs of complications. I will contact my Bloom Metabolics care team via the patient portal with any questions before injecting."

If you have questions about any part of these instructions, contact your care team BEFORE completing the acknowledgment and BEFORE injecting.

Clinical References

Testosterone Cypionate Injection, USP (NDA 085635). Current FDA labeling at time of protocol drafting.

- AUA Clinical Guideline: Evaluation and Management of Testosterone Deficiency (2018, updated 2024).
- Endocrine Society Clinical Practice Guideline: Testosterone Therapy in Men with Hypogonadism (2018).
- CDC Guidelines: Injection Practices — Intramuscular Injections (current).
- Note: FDA removed the cardiovascular black box warning from testosterone products in 2025 following the TRAVERSE (Testosterone Replacement Therapy for Assessment of Long-term Vascular Events and Efficacy ResponSE in Hypogonadal Men) trial results.

This protocol is governed by SOP-004 (Patient Self-Administration Injection Education).

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